# Bhadra Korukonda

Software Engineer – AI systems
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Detail-oriented Computer Science student with hands-on experience in software engineering, system automation, and AI-driven tooling. Developed scalable backend systems, CI/CD pipelines, and data-driven simulation frameworks integrating Python, cloud infrastructure, and monitoring tools. Skilled in performance optimization, distributed testing, and full-stack deployment.

## SKILLS

**Programming Languages:** Python, Java, JavaScript
**Libraries/Frameworks:** PyTorch, scikit-learn, NumPy, Flask, React
**Tools/Platforms:** FastAPI, Flask, REST APIs, Git, AWS, Docker, Kubernetes, Prometheus, Grafana
**Databases:** PostgreSQL, MongoDB, SQL

## PROFESSIONAL EXPERIENCE

**Research Centre Imarat(RCI), DRDO**
**Data Scientist Intern |** June 2024 – Aug 2024

- Automated Git workflows with Python, cutting version control overhead by **30%**.
- Developed statechart validation tools using OOP and tests, boosting accuracy by **25%**.
- Streamlined pipelines with SQL and Git CI, accelerating debugging and deployment.

**IIT Hyderabad**
**Research Intern |** Python, PyTorch Geometric, Pandas, Geopy, GRU/LSTM, GNNs (GCN, GAT, EGAT) May 2025 – Aug 2025

- Implemented a hybrid Edge-Aware Graph Attention Network (EGAT-GRU) to forecast PM2.5 levels across Delhi using spatio- temporal air quality data.
- Encoded wind direction and geodesic distance as edge features, enabling wind-informed pollutant transport learning and dynamic spatial attention.
- Achieved state-of-the-art performance (RMSE = 16.72, MAE = 10.05), outperforming GAT, GCN, and physics-based baselines for real-time pollution prediction.

## PROJECTS

**Cloud Automation Pipeline |** Docker, AWS, Grafana, Prometheus, CI/CD, Pytho**n**

- Engineered an automated **CI/CD pipeline** to streamline build, test, and deployment workflows using Docker and AWS.
- Integrated **Grafana dashboards** to visualize system metrics and deployment health in real time.
- Automated regression testing and environment provisioning, improving deployment reliability and reducing setup time by 40%.
- Implemented container orchestration principles to simulate **distributed testing environments**, mirroring real-world load scenarios.

**RelFuse MVP |** Pytorch, DGL, Transformers, GraphSAGE, FastAPI

- Developed a hybrid relational-feature fusion model combining GNN embeddings with transformer attention for structured data reasoning.
- Conducted feature ablation and hyperparameter tuning, achieving +18 % classification accuracy on benchmark graph datasets.
- Built modular inference APIs to serve predictions and visualizations, bridging research models with deployable analytics tools.

**Automated Resume Ranking System using NLP and MCDM | Python, Flask, spaCy, scikit-learn ,PyMCDM, Tesseract OCR**

- Developed an AI-driven resume evaluation system integrating Natural Language Processing (NLP) and Multi-Criteria Decision-Making (MCDM) for automated, explainable candidate ranking.
- Extracted and analyzed resume data (skills, education, experience, projects) using spaCy, NLTK, and Tesseract OCR to generate structured features for scoring.
- Implemented TOPSIS/AHP algorithms to compute weighted candidate scores, improving shortlisting efficiency by ~40% and reducing human bias.
- Built a Flask-based API and dashboard enabling HR users to upload resumes, adjust evaluation weights, and visualize ranked results in real time.
- Ensured modular design for scalability and transparency, allowing independent updates to NLP or MCDM components.

## EDUCATION

**Bachelor of Technology (CS Engg), Woxsen University**, GPA: 3.0/4.0 Aug 2022 – July 2026
**Narayana Jr College(SSC), Hyderabad**, Percentage: 87.1% June 2020– May 2022
**Chinmaya Vidyalaya(CBSE)**,Hyderabad Percentage: 90% May 2008 – June 2020